

Julio Cesar Aldana Almanza

Instrumented & Monitored Cloud Service





Agenda

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- Architecture & Instrumentation
 - Alerting system
 - Live Demo
 - Incidence Response
 - Learnings & Improvements

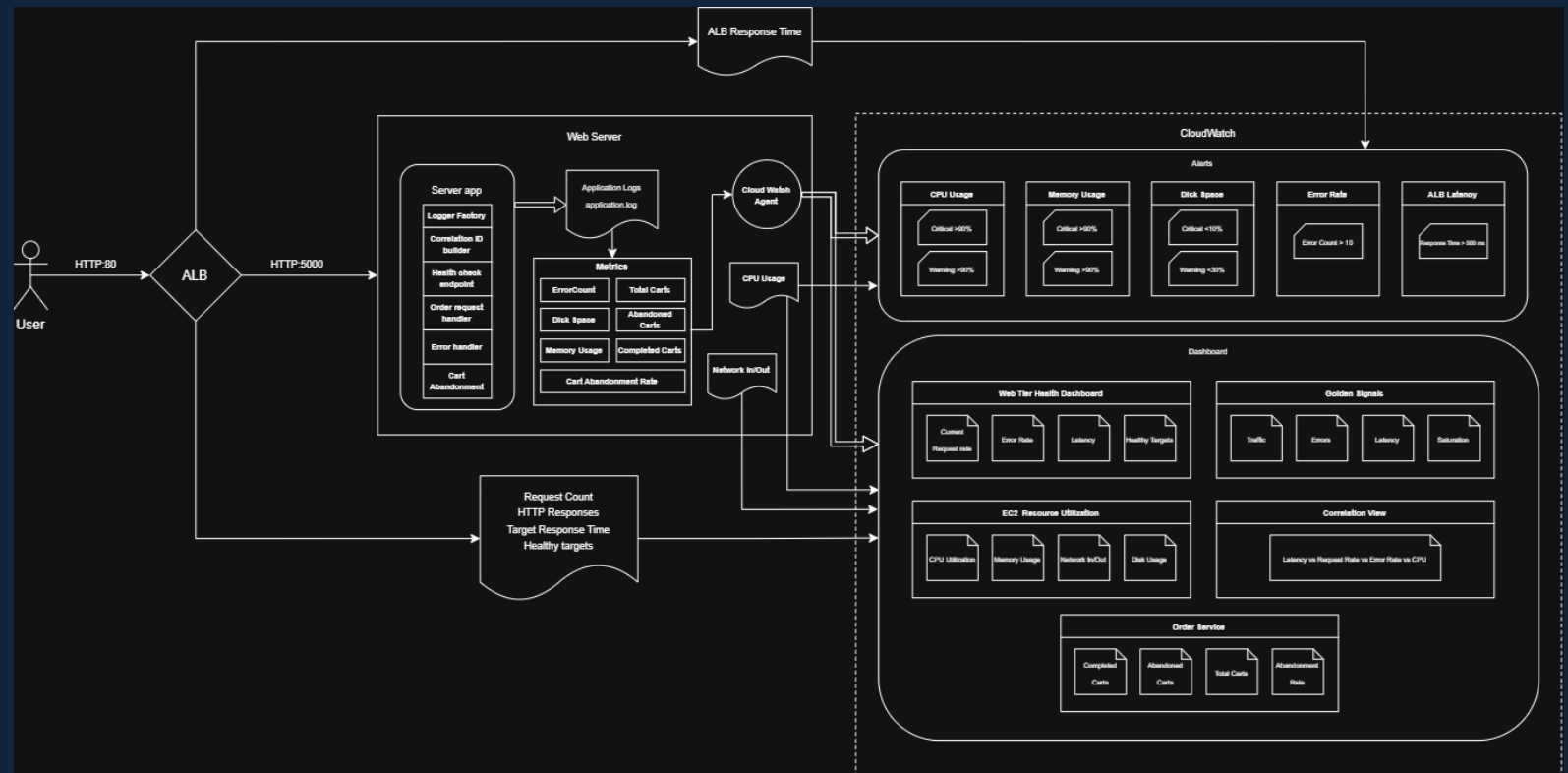


Architecture & Instrumentation

Instrumented & Monitored Cloud

Architecture

- Web Server
 - Logger Factory
 - application.log
- Application Load Balancer
- CloudWatch Agent
- CloudWatch Alerts
- CloudWatch Dashboard



Instrumentation

- Correlation ID
- Cloud Watch Agent
- Filter Logs

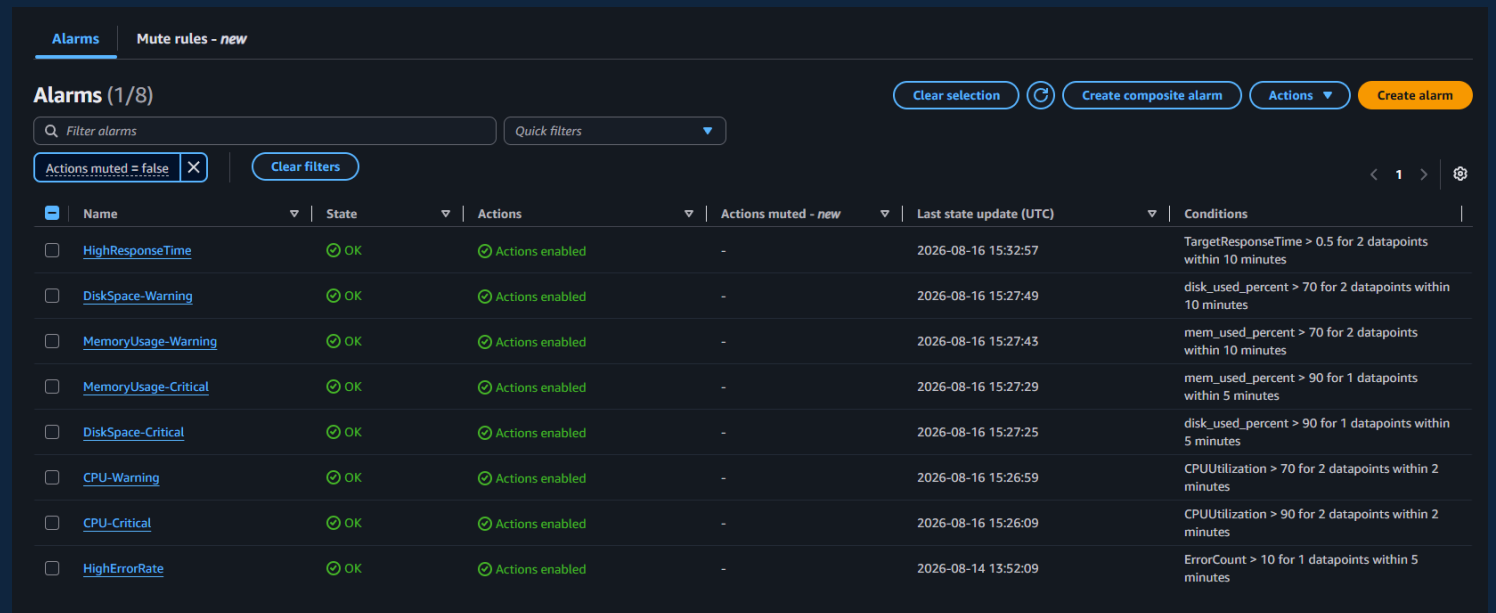
```
{  
  "event": "cart_abandonment_metric",  
  "metric_name": "CartAbandonmentRate",  
  "metric_value": 40.0,  
  "total_carts": 10,  
  "abandoned_carts": 4,  
  "completed_carts": 6,  
  "active_carts": 0  
}
```

Technical metrics	Business Metrics
Memory Usage	Total Carts
Disk Space	Completed Carts
Application Error Rate	Abandoned Carts
	Abandonment Cart Rate

EC2 Metrics	ALB Metrics
CPU Usage	Target Response Time
Network In/Out	Request Count
	Healthy Targets
	HTTP Code Responses

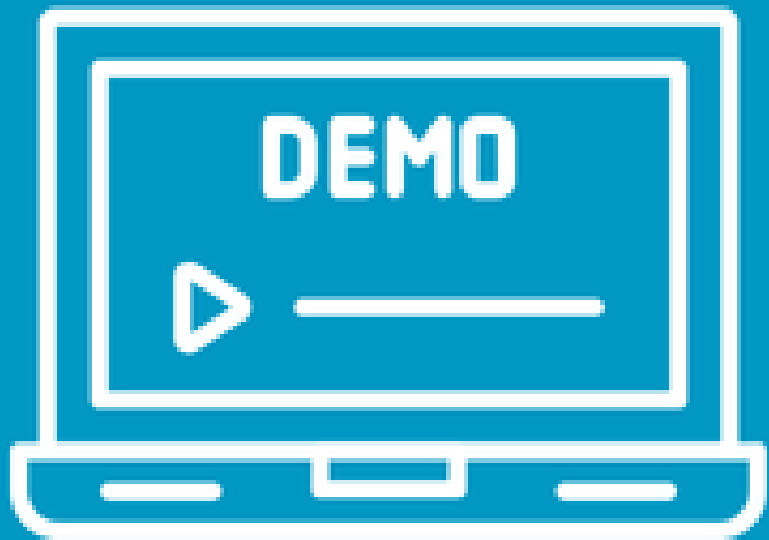
Alerting System

- CPU Usage (Tiered Alerting)
 - Critical >90%
 - Warning >70%
- Memory Utilization (Tiered Alerting)
 - Critical >90%
 - Warning >70%
- Disk Space (Tiered Alerting)
 - Critical >90%
 - Warning >70%
- Application Error Rate
- Application Load Balancer Response Time



The screenshot displays the 'Alarms' section of a monitoring system. At the top, there are tabs for 'Alarms' and 'Mute rules - new'. Below the tabs, the title 'Alarms (1/8)' is shown. A search bar labeled 'Filter alarms' and a 'Quick filters' dropdown are present. A filter button indicates 'Actions muted = false'. A 'Clear filters' button is also visible. On the right, there are buttons for 'Clear selection', 'Create composite alarm', 'Actions', and 'Create alarm'. The main content is a table with columns: Name, State, Actions, Actions muted - new, Last state update (UTC), and Conditions. The table lists eight alarms, all with a state of 'OK' and 'Actions enabled'.

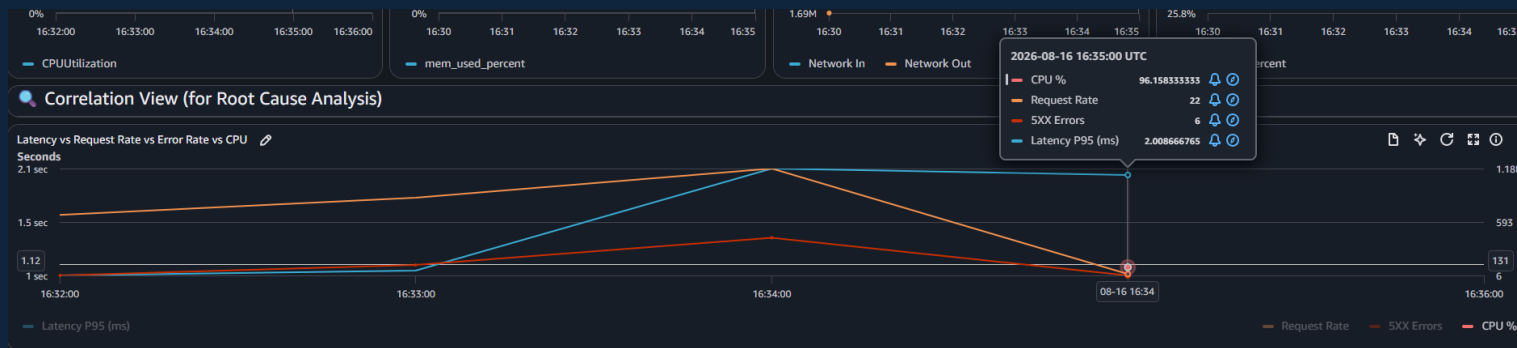
Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
HighResponseTime	OK	Actions enabled	-	2026-08-16 15:32:57	TargetResponseTime > 0.5 for 2 datapoints within 10 minutes
DiskSpace-Warning	OK	Actions enabled	-	2026-08-16 15:27:49	disk_used_percent > 70 for 2 datapoints within 10 minutes
MemoryUsage-Warning	OK	Actions enabled	-	2026-08-16 15:27:43	mem_used_percent > 70 for 2 datapoints within 10 minutes
MemoryUsage-Critical	OK	Actions enabled	-	2026-08-16 15:27:29	mem_used_percent > 90 for 1 datapoints within 5 minutes
DiskSpace-Critical	OK	Actions enabled	-	2026-08-16 15:27:25	disk_used_percent > 90 for 1 datapoints within 5 minutes
CPU-Warning	OK	Actions enabled	-	2026-08-16 15:26:59	CPUUtilization > 70 for 2 datapoints within 2 minutes
CPU-Critical	OK	Actions enabled	-	2026-08-16 15:26:09	CPUUtilization > 90 for 2 datapoints within 2 minutes
HighErrorRate	OK	Actions enabled	-	2026-08-14 13:52:09	ErrorCount > 10 for 1 datapoints within 5 minutes



Live Demo

Incidence Summary & Solution

- **Incident:** API latency increased from 200ms (P95) to 2,000ms (P95)
- **Impact:** 250 users experienced slow page loads, 35% error rate
- **Duration:** 60 minutes (17:00-18:00 UTC, Aug 14, 2024)
- **Root Cause:** Unexpected traffic/application workload



Rate (Traffic)

- Normal: 25 req/min
- Incident: 632 req/min - See Health Dashboard Error Rate
- Finding: 10x traffic increase

Errors

- Normal: 0.1%
- Incident: 35.0% during the incident - See Golden Signals Error Rate
- Finding: 70x error increase

Duration (Latency)

- Normal P95: 200ms
- Incident P95: 2,000ms
- Finding: 10x latency increase

CPU

- Utilization: 95% (abnormal)
- Saturation: Abnormal
- Errors: Timeout errors

Learnings & Improvements





Learnings

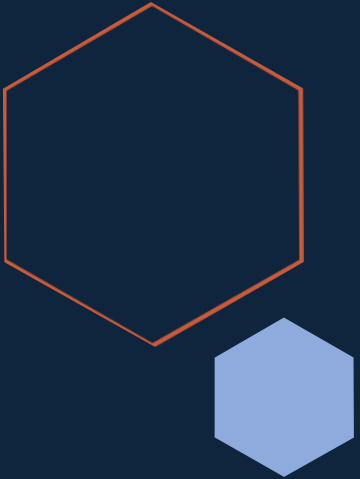
1. Create logs with sufficient data and identified to track relevant information for troubleshooting.
2. Filter log info to monitor the server and create alerts to perform actions immediately.
3. Functionality of a widget, how it is populated, sample frequency and display type.
4. Use of RED/USE Methodology to find the root cause.

Improvements

1. Redesign logging of cart requests to get proper values for active carts to be displayed in the dashboard.
2. Enable duration metric to be filtered in the log insights.
3. Increase the number of business metrics and eventually create a dashboard for business metrics
4. Add cart storage using DynamoDB or another datastore.



Q&A



A decorative pattern of hexagons in various shades of blue, orange, and white, arranged in a honeycomb-like structure on the left side of the slide.

Thank you

Julio Cesar Aldana Almanza